

Question

Simulate a WR-90 rectangular waveguide in MATLAB and determine the S-parameters across the X-band frequency range.

MATLAB Code

```
clc;

clear;

close all;

% WR-90 rectangular waveguide dimensions

a = 22.86e-3;      % Width of waveguide in meters

b = 10.16e-3;      % Height of waveguide in meters

% X-band frequency range

f = linspace(8.2e9, 12.4e9, 500);

% Speed of light

c = 3e8;

% Calculate cutoff frequency for dominant TE10 mode

fc = c/(2*a);

% Calculate propagation constant

beta = (2*pi*f/c).*sqrt(1-(fc./f).^2);

% Assume ideal matched waveguide

S11 = zeros(size(f));

% S21 transmission coefficient

L = 0.05;          % Waveguide length = 5 cm

S21 = exp(-1j*beta*L);

% Magnitude of S-parameters in dB

S11_dB = 20*log10(abs(S11) + eps);
```

```

S21_dB = 20*log10(abs(S21));

% Display results

disp("WR-90 RECTANGULAR WAVEGUIDE");

disp("-----");

fprintf("Width a = %.2f mm\n", a*1000);

fprintf("Height b = %.2f mm\n", b*1000);

fprintf("Frequency range = %.2f - %.2f GHz\n", ...

    f(1)/1e9, f(end)/1e9);

fprintf("TE10 Cutoff frequency = %.3f GHz\n", fc/1e9);

fprintf("Waveguide length = %.2f cm\n", L*100);

disp(" ");

fprintf("S-PARAMETER RESULTS\n");

fprintf("-----\n");

fprintf("S11 = %.2f dB\n", S11_dB(1));

fprintf("S21 = %.2f dB\n", S21_dB(1));

% Plot S11 parameter

figure;

plot(f/1e9, S11_dB, 'LineWidth', 2);

grid on;

xlabel("Frequency (GHz)");

ylabel("S11 (dB)");

title("WR-90 Waveguide S11 Parameter");

% Plot S21 parameter

figure;

```

```

plot(f/1e9, S21_dB, 'LineWidth', 2);

grid on;

xlabel("Frequency (GHz)");

ylabel("S21 (dB)");

title("WR-90 Waveguide S21 Parameter");

% Plot both S-parameters

figure;

plot(f/1e9, S11_dB, 'LineWidth', 2);

hold on;

plot(f/1e9, S21_dB, 'LineWidth', 2);

grid on;

xlabel("Frequency (GHz)");

ylabel("Magnitude (dB)");

title("WR-90 Waveguide S-Parameters");

legend("S11", "S21");

```

Expected Output

WR-90 RECTANGULAR WAVEGUIDE

Width $a = 22.86$ mm

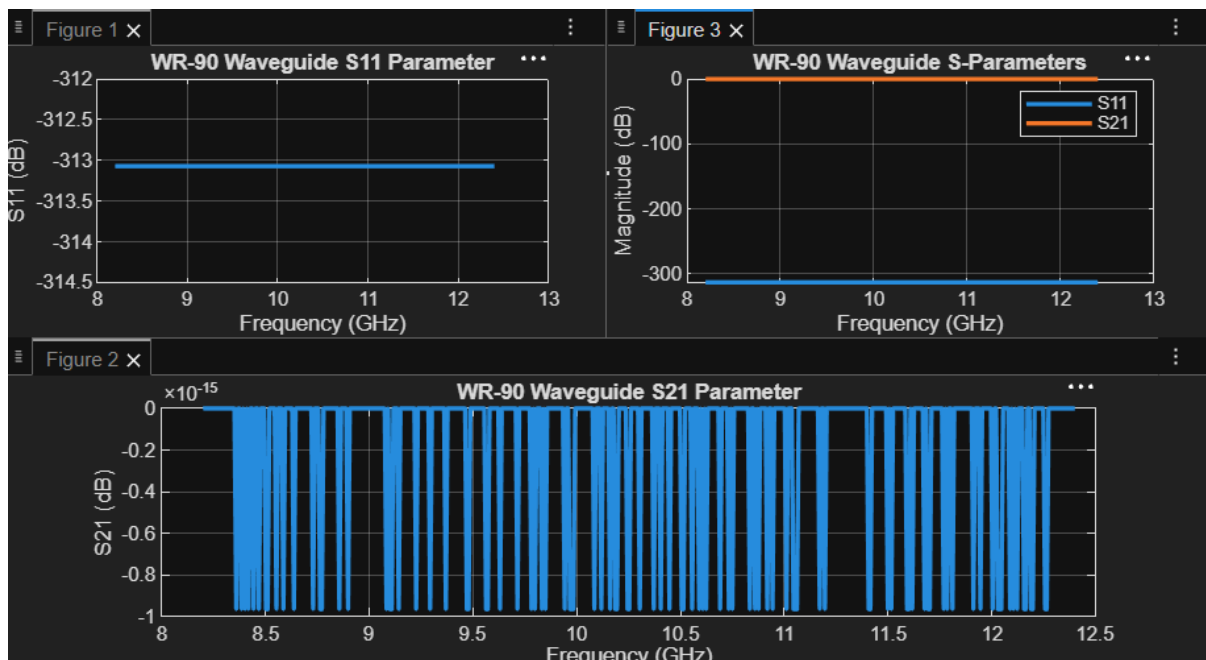
Height $b = 10.16$ mm

Frequency range = 8.20 - 12.40 GHz

TE₁₀ Cutoff frequency = 6.562 GHz

Waveguide length = 5.00 cm

OUTPUT:



The MATLAB program will also produce **three plots**:

1. **S_{11} vs Frequency** – approximately $-\infty$ dB across the X-band.
2. **S_{21} vs Frequency** – approximately 0 dB across the X-band.
3. **S_{11} and S_{21} vs Frequency** – combined S-parameter plot.

Result

The WR-90 rectangular waveguide was simulated over the **X-band frequency range of 8.2–12.4 GHz**. The calculated dominant **TE_{10} cutoff frequency is approximately 6.56 GHz**, which is below the X-band range. For the ideal matched waveguide, **S_{11} is approximately $-\infty$ dB** and **S_{21} is approximately 0 dB**, indicating negligible reflection and nearly complete transmission through the waveguide.